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VATTAMALAIPALAYAM, N.G.G.O. COLONY POST, COIMBATORE – 641 022

Batch No:2CS56

20CS56 – Mini Project I

"Development of AI-Driven Social Platform with FAQ Support"

Type of Project: Application / Society Based

Presented by
Sivabalan A – 71812401234
Sowmyasree S - 71812401235
Pitambar Yadav - 71812401266

Supervised by
Mr.S .Suresh Kumar
(Assistant Professor / Sl.G)

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Contents

S.No	Description	Slide Number
1	Introduction	3
2	Existing technology of the project	4
3	Problem identification, scope and objective	5
4	Methodology	6
5	Timeline	7
6	Proposed Outcome and budget	8
7	References	9

Introduction

- In many colleges, event updates and announcements are conveyed manually via students/class Representatives.
- Social media is not specifically built for college needs—thus missing relevant academic features.....
- There is no dedicated platform that provides access to past year question papers and academic materials in an organized way. Our chatbot is enhanced to instantly share links to past year question papers on request.
- Some institutions use Learning Management Systems (LMS) like Moodle or Google Classroom, but they do not integrate chatbot-based FAQs, social interaction, or student analytics in one platform.
- Our project addresses these gaps through a unified web platform with Firebase, AI-powered FAQ Chatbot, and NLP-based text analyzer.

Current technology of the project (5 Slides)

- Personality-Adapted Chatbots: Chatbots that adjust their tone, language, and responses based on user personality traits to create better engagement.
- Educational Advising Bots: Chatbots used as virtual academic advisors that can respond to student queries about curriculum, scheduling, and institutional policies.
- User Behavior Analysis: Systems that evaluate how users interact with different chatbot personalities to improve UX and retention.
- Emotionally Congruent Interfaces: Development of chatbot interfaces and response patterns that feel more “human,” improving user satisfaction and interaction depth.

Source: “Assessing the Impact of Chatbot-Human Personality Congruence on User Behavior: A Chatbot-Based Advising System Case”, Mohammad Amin Kuhail, Mohamed Bahja, Ons Al-Shamaileh, Justin Thomas, Amina Alkazemi, and Joao Negreiros, IEEE Access, Vol. 12, No. –, pp. –, 2024. (Scopus Indexed)

Current technology of the project (5 Slides)

- LLM-based Chatbots (like GPT): Used to answer student queries related to specific academic domains such as databases and information systems.
- Conversational Interfaces: Provides intuitive communication between users (students) and systems through chat-style interaction.
- Domain-Specific Knowledge Bases: Integrating structured academic data into chatbot systems for accurate answers.
- Feedback Loops for Improvement: Using real student interactions to continuously improve chatbot performance and relevance.
- Educational Content Integration: Ability to explain concepts, provide examples, and clarify doubts similar to human tutors.
- Scalability for Large Student Populations: Designed to handle queries from many users simultaneously with consistent performance.

Source: "*An LLM-Driven Chatbot in Higher Education for Databases and Information Systems*" by Alexander Tobias Neumann, Yue Yin, Sulayman Sowe, Stefan Decker, and Matthias Jarke, IEEE Transactions on Education, Vol. 68, No. 1, pp. 103–112, February 2025, Scopus Indexed.

Current technology of the project (5 Slides)

- Use of Natural Language Processing (NLP) to analyze student-generated content on academic forums (e.g., Reddit).
- Implementation of machine learning models to classify stress-related vs. non-stress posts.
- Preprocessing techniques like stopword removal, lemmatization, and tokenization to handle noisy student-written text.
- Use of open social platforms to monitor mental health signals among students using AI-based tools.
- Integration of emotion detection (e.g., stress, anxiety, frustration) through trained classifiers (e.g., SVM, Logistic Regression).

Source: "Detection and Analysis of Stress-Related Posts in Reddit's Academic Communities", Nazzere Oryngozha, Pakizar Shamoi, and Ayan Igali, IEEE Access, Vol. 12, No. –, pp. –, 2024. (Scopus Indexed)

Current technology of the project (5 Slides)

- Rule-Based Chatbots: Many current systems use predefined question-answer sets, which are limited in flexibility and scope.
- Dialogue Management Systems: Existing learning bots employ basic dialog flow managers like Dialogflow or Rasa to handle structured conversations.
- Natural Language Processing (NLP): Standard NLP techniques such as tokenization, intent recognition, and entity extraction are used to understand student inputs.
- Pretrained Language Models: Use of foundational models (like BERT or GPT variants) to improve language understanding in educational bots.
- Text-Based Interface: Existing platforms usually rely on simple text-based interaction without multimedia or emotional intelligence.
- Many systems rely on limited dialog platforms (like Dialogflow), which restrict dynamic conversation handling.

Source: "*A Conversational AI Bot for Efficient Learning: A Prototypical Design,*" Ahmed Elragal, Ali Ismail Awad, Ingemar Andersson, and Jörgen Nilsson, IEEE Access, Vol. 12, pp. 124567–124580, October 2024. (Scopus Indexed)

Current technology of the project (5 Slides)

- AI-driven Learning Analytics: AI is used to analyze student behavior and participation in virtual courses to improve academic outcomes and retention rates.
- AI Chatbots for Student Support: Virtual assistants answer frequently asked questions and guide students through course materials or administrative processes.
- Sentiment Analysis and Engagement Monitoring: AI tools analyze textual and behavioral data to identify disengaged or struggling students.
- Automated Notifications & Reminders: AI systems track participation and send alerts or motivational messages to improve consistency in course engagement.

Source: "Application of Artificial Intelligence in Online Education: Influence of Student Participation on Academic Retention in Virtual Courses", William Villegas-Ch, Joselin García-Ortiz, and Santiago Sánchez-Viteri, IEEE Access, Vol. 12, No. –, pp. –, 2024. (Scopus Indexed)

Problem identification, scope and objectives

Problem Identification:
Students, especially new or incoming ones, face scattered platforms for academics, events, and communication. The SREC website is hard to navigate, past papers require manual library search, and there's no real-time help. There's also no dedicated social space to post content, connect, or see club event updates — making campus life less engaging and organized.

Scope:
To build a secure, college-only platform for SREC where students can post content, view club events, access past papers, and get real-time help through an AI chatbot — all in one place to boost academic and social engagement.

Objectives:

- Build a secure, centralized platform for all SREC students — especially newcomers — to access academic and campus information easily.
- Enable students to post content, view club events, and connect socially, with a sentiment analyzer to understand post tone.
- Provide quick access to past year question papers and real-time help through an AI-powered FAQ chatbot.
- Ensure easy data updates and user authentication to keep the platform reliable and safe.

Methodology to be adopted

- The platform will be developed using modern full-stack web technologies: HTML, CSS, JavaScript, Firebase, and Firestore for backend data management.
- User authentication and post management will be handled using Firebase Authentication and Firestore Database.
- Posts created by students can include text and images, which will be uploaded using Imgbb API.
- An AI-powered chatbot will be integrated using Google Apps Script and Google Sheets as the knowledge base for answering FAQs.
- Gemini API is integrated to analyze the sentiment of each post dynamically using typewriter animation effects for better UX.
- Admins or club reps will have privileges to post upcoming events or important announcements.

Timeline of the Project and Budget

Timeline

- **month 1**: Plan features, gather requirements, rough UI sketches
- **month 2–3**: Design login, post, chatbot screens (UI/UX)
- **month 3-4**: Firebase setup, frontend + backend dev, chatbot link to Sheets
- **month 4–5**: Add sentiment analysis (Gemini API), test chatbot replies
- **month 6-7**: Bug fixes, performance checks, UI polish
- **month 7-8**: Final deployment, documentation

Estimated Cost

Item	Cost
Firebase	Free Tier
Google Sheets & Script	Free
Gemini API	Free (Basic)
Domain/Hosting *(optional)*	₹800–₹1000/year
Design Assets	Free

Proposed Outcome of the Project

- A user-friendly web-based college social networking platform exclusively for SREC students.
- Enables students to post thoughts, share images, and interact within a secure campus-only environment.
- Integrated AI-powered chatbot for answering frequently asked questions (FAQs), including:
- General college queries (fees, timings, departments, SREC website summary).
- Club-related queries and event information.
- Access to past-year question papers via chatbot & Google Drive links.
- Sentiment analyzer integrated with each post to assess and display post emotions (positive, neutral, negative).
- Future-ready design that can support:
- Club event announcements.

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